TEST_COVERAGE_PLAN

Helix OTA — Emulation Test-Fabric — Test-Coverage Plan

Field	Value
Revision	1
Last modified	2026-06-10T14:10:00Z
Status	active — plan
Status summary	How EVERY supported test type (§11.4.27) maps onto the fabric tiers in `DESIGN.md`, the anti-bluff captured-evidence shape per type (§11.4.5/§11.4.69/§11.4.107), and how 100%-feature×flow×edge-case coverage is MEASURED and ratcheted (§11.4.50) — never asserted.
Related	DESIGN.md; ROADMAP.md; SCHEMA.sql; `docs/research/main_specs/1.0.0-mvp/qa/coverage_ledger.md`

1. Test-type × tier × evidence matrix

Test type	Primary tier(s)	Anti-bluff captured evidence (the PASS must cite)
unit	host (Go/Kotlin)	`go test`/Gradle transcript; mocks permitted ONLY here (§11.4.27)
integration	T0 + `pkg/fabric` registry	real-Postgres parity (`-tags integration`, podman-booted PG, 0 skips); request/response transcript
e2e	T0/T1/T2/T3	full register→update-check→telemetry→delta→rollout→recall transcript + server-side history cross-check
full-automation	T1/T2/T3	

Test type	Primary tier(s)	Anti-bluff captured evidence (the PASS must cite)
		self-driving (no human in loop, §11.4.98); ADB/`update_engine_client` state + UI capture
security	Tcp + T0	authz/ownership/injection/trust-boundary probes scored on real HTTP status (e.g. recall+telemetry probes 28/0)
ddos / scaling	Tcp (Firecracker/ Kata)	measured req throughput + shed/served counts + recovery (§11.4.85)
chaos	T2 + Tcp + `pkg/fabric`	fault-injection (corrupt-slot→auto-rollback; lease loss; PG SIGKILL) + recovery trace (§11.4.85)
stress	T0 fleet + registry	N-device/N-lease sustained load, p50/ p95/p99 latency, 0 lost updates
performance / benchmarking	T0 + Tcp	`testing.B` ns/op + alloc; memory-vs-pgx percentile sweep
ui / ux	T1 (AVD)	uiautomator/ADB-driven journey + screen capture; §11.4.107 liveness oracle where pixels matter
Challenges (HelixQA)	all tiers	bank challenge dispatching to the real test, scored on a non-empty evidence ledger (`run_bank.sh` + canonical engine)
autonomous QA session (§11.4.27)	all tiers	HelixQA session driving every registered bank with captured per-check evidence

2. Real A/B fidelity boundary (FACT, §11.4.6)

Real `update_engine` A/B + AVB/dm-verity + auto-rollback evidence is produced ONLY by **T2 (Cuttlefish)** and **T3 (real RK3588)**. T0 reports the lifecycle but fakes the flash; T1 (stock AVD) runs Android userspace but real-A/B-apply under AVD is `UNCONFIRMED:` pending the P1 GSI-A/B empirical test. The coverage ledger records this boundary so no row claims A/B fidelity it lacks.

3. 100% coverage — measured, not claimed

- **Inventory.** Every feature × flow × use-case × edge-case is a row in the §11.4.25 coverage ledger, each cross-referenced to its test(s), tier(s), and the six §11.4.25 invariants.
- **Measurement.** Go: `go test -coverprofile` per package (≥90% safety floor already enforced on the critical handlers). Fabric/e2e: the ledger row is COVERED only when its real test exists AND passed with captured evidence in the last cycle — a declared-but-unrun row is a §11.4 bluff (the exact HelixQA-bank gap the `run_bank.sh` runner closed).

- **Ratchet (§11.4.50).** The covered/total ratio is a gate that only moves up; a regression in the ratio FAILs the release sweep. Edge-case discovery (§11.4.118) feeds new rows.
- **No skip-as-pass.** A tier that cannot run on a host SKIPs-with-reason (§11.4.3) and the ledger shows the gap honestly; it never counts a SKIP as coverage.

4. Per-change obligation (§11.4.146 / §11.4.135)

Every fix/feature on the fabric: reproduce-first RED on the broken artifact → same-test GREEN on the fix → extend across the functionality's case-space (valid/invalid/boundary/concurrent/chaos/ topology) → register a permanent regression guard → add the HelixQA challenge → update the ledger with measured coverage. Independent-agent review (§11.4.125/§11.4.142) confirms before merge.